

HEAT TRANSFER

III B. Tech. II Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME31	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OVERVIEW:

This course will provide knowledge about application of conduction, convection and radiation heat transfer concepts to different practical applications.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Demonstrate the basic modes of heat transfer and compute temperature distribution on steady and unsteady state heat conduction.
2. Analyse the heat transfer through variable thermal conductivity, system with heat generation and extended surfaces.
3. Interpret and analyse the free and forced convection heat transfer.
4. Comprehend the concepts of Boiling, Condensation and Heat exchangers.
5. Explain the principles of radiation heat transfer and apply the radiation laws on black and gray bodies.

UNIT-I	INTRODUCTION
Modes and mechanisms of heat transfer, Basic laws of heat transfer, Applications of heat transfer. Fourier Law of conduction, general heat conduction equations in Cartesian, Cylindrical and Spherical coordinates. Different forms of the field equation, steady, unsteady and periodic heat transfer – Initial and boundary conditions.	

UNIT-II	ONE DIMENSIONAL HEAT CONDUCTION
Heat transfer through homogeneous slabs, hollow cylinders and spheres, Overall heat transfer coefficient, Electrical analogy and Critical radius of insulation. Systems with internal heat generation. Extended surfaces (Fins), Long, Short and insulated tips. Significance of Biot and Fourier Numbers, Chart solutions of transient conduction systems	

UNIT-III	CONVECTION
CONVECTIVE HEAT TRANSFER: Concepts of Continuity, Momentum and Energy Equations. Dimensional analysis – Buckingham’s Pi Theorem – Application for developing non-dimensional correlation for convective heat transfer. FORCED CONVECTION: External Flows, Concepts of hydrodynamic and thermal boundary layer and use of empirical correlations for Flat plates and Cylinders. Internal Flows, Concepts about Hydrodynamic and Thermal Entry Lengths, use of empirical correlations for Horizontal Pipe Flow and annulus flow. FREE CONVECTION: Development of Hydrodynamic and thermal boundary layer along a vertical plate, Use of empirical relations for Vertical plates and pipes.	

UNIT-IV	BOILING & CONDENSATION, HEAT EXCHANGERS
BOILING & CONDENSATION: Regimes of Pool boiling and Flow boiling, Critical heat flux, Calculations on Nucleate Boiling. Film wise and drop wise condensation, Nusselt’s theory of condensation on a vertical plate – Film condensation on vertical and horizontal cylinders using	

empirical correlations. HEAT EXCHANGERS: Classification of heat exchangers, overall heat transfer Coefficient and fouling factor, Concepts of LMTD and NTU methods, Problems using LMTD and NTU methods.	
UNIT-V	RADIATION HEAT TRANSFER
Emission characteristics, Laws of black-body radiation, Irradiation, Total and monochromatic quantities, Laws of Planck, Wien, Kirchhoff, Lambert, Stefan and Boltzmann, Heat exchange between two black bodies, concepts of shape factor, Emissivity, heat exchange between grey bodies, radiation shields.	
Text Books:	
1. Yunus A. Cengel, Heat Transfer a Practical Approach, 4th Edition, Tata McGraw hill education (P) Ltd, New Delhi, India, 2012. 2. R. C. Sachdeva, Fundamentals of Engineering, Heat and Man Transfer, 3rd Edition, New Age, New Delhi, India, 2012.	
Reference Books:	
1. J.P. Holman, Heat Transfer (SI Units), 10th Edition, Tata McGraw hill education (P) Ltd, New Delhi, India, 2012. 2. P. S. Ghoshdastidar, Heat Transfer, 2nd Edition, Oxford University Press, New Delhi, India, 2012. 3. Incropera, Dewitt, Fundamentals of Heat Transfer, 6th Edition, John Wiley, UK, 2012.	